P-S-591H
February 10, 1976
SUPERSEDING
Fed. Spec. P-S-591G
December 28, 1965

FEDERAL SPECIFICATION

SOAP, LAUNDRY, BAR

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of laundry soap (see 6.1).

1.2 Classification. The laundry soap covered by this specification shall be of one grade and shall be of the following sizes, as specified (see 6.2):

Size 1 - 1 pound bar.
Size 2 - 1/2 pound bar.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-1163 - Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax and Resin Impregnated.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 224 - Closing, Sealing and Reinforcing of Fiberboard Shipping Boxes, General Methods For.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials:

ASTM D 460 - Soap and Soap Products, Sampling and Chemical Analysis Of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Composition. The laundry soap shall be thoroughly saponified and its composition shall conform to the requirements specified in table I when tested as specified in 4.4. The titer of the mixed fatty acids prepared from the soap shall be not lower than 29°C.

TABLE I. Analytical requirements

Characteristic	Minimum	Maximum
Moisture and matter volatile at 105°C, percent	---	36.0
Sum of free alkali or free acid, total matter insoluble in alcohol and sodium chloride, percent	---	11.0
Free alkali, calculated as sodium hydroxide (NaOH), percent	---	0.2
Free acid, calculated as oleic acid, percent	---	0.5
Matter insoluble in water, percent	---	1.0
Chloride, calculated as sodium chloride (NaCl), percent	---	1.0
Alkaline salts, calculated as sodium oxide (Na_2O) percent	1.0	6.5
Rosin, percent	---	25.0
Anhydrous soap, percent	52.0	---

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3.1.1 Computation. To determine compliance with the composition requirements of table I, the moisture and volatile matter content shall be expressed as determined on the soap and the test results of all other ingredients shall be expressed on the basis of a soap containing 36 percent moisture and volatile matter. To express these results on a 36 percent moisture and volatile matter basis, the analytical test results are multiplied by a factor F, where:

$$F = \frac{100 - 36}{100 - M}$$

M = Average percent moisture and volatile matter at 105°C of the lot as determined from the composite.

3.2 Form. The laundry soap shall be furnished as firm bars.

3.3 Weight. The average bar weight for size 1 shall be 16 ± 1/2 ounce. The average bar weight for size 2 shall be 8 ± 1/4 ounce. 1/

3.4 Dimensions. Size 1 laundry soap shall be 4-1/2 by 2-3/4 by 2-1/4 inches ± 3/8 inch tolerance in each dimension. Size 2 laundry soap shall be 4-1/2 by 2-3/4 by 1-1/8 inches ± 1/4 inch tolerance in each dimension. 1/

3.5 Color. A freshly cut surface of the laundry soap shall have a uniform color (a slight variation in color occurring within 1/4 inch or any edge shall be disregarded). The darkest portion of a freshly cut bar shall not be darker than color No. 30266 of Fed. Std. No. 595. 1/

3.6 Odor. The laundry soap shall be free of objectionable odors at room temperature (70° to 80°F) and in a solution of the soap in water at 130° to 140°F. An objectionable odor, for this specification, is defined as a rancid, fishy, or horsy odor.

3.7 Workmanship. The laundry soap shall be uniform in appearance and texture. Bars shall be free of foreign matter and unduly sticky surfaces. The laundry soap shall conform to the levels of quantity established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling, except for examination of preparation for delivery, for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot shall consist of all laundry soap of the same size from one manufacturing batch.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

1/ These requirements shall be determined on the bar without any moisture adjustment.

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4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.5. The lot size shall be expressed in units of bars of laundry soap of one size for the examinations in 4.3.2, and in units of shipping containers for the examinations in 4.3.3 and 4.3.4.

4.3.2 Examination of laundry soap for defects in form, dimensions, color, odor, and workmanship. The sample unit for this examination shall be one bar of soap.

Examine	Defect
Form	Not bar; not firm (crumbly).
	Not in one piece.
Dimensions	Not specified dimensions.
Color	Color not uniform.
	Darker than color chip No. 30266.
Odor	Rancid, fishy, or horsy.
Workmanship	Not uniform in appearance and texture.
	Presence of foreign matter.
	Unduly sticky.

4.3.3 Examination for defects in average weight per bar of laundry soap. The sample unit for this examination shall be the net contents of one shipping container. The average weight per bar shall be within the specified weight. The net contents of no less than 10 shipping containers shall be weighed. If the lot consists of less than 10 shipping containers, the net contents of all shipping containers shall be weighed.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the packaging, packing and marking requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

TABLE II. Examination of preparation for delivery

Examine	Defect
Container	Not as specified.
Content	Not as specified.
Marking	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Material	Component missing or damaged.
Workmanship	Bulging or distortion of containers.

4.3.4.1 Examination of closure, waterproofing, and banding of container. When the shipping container is required to comply with PPP-B-636, the examination for defects in closure, waterproofing, or banding shall be in accordance with the appendix of the box specification.

4.3.5 Inspection levels and AQL's for examinations. The inspection levels, for determining the sample size, and AQL's expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2	I	2.5
4.3.3	S-3	-
4.3.4	S-4	2.5

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4.4 Testing of the laundry soap. The methods of testing listed in table III shall be followed for each lot. American Society for Testing and Materials (ASTM) Test Methods shall be used as indicated in Table III. The lot size shall be expressed in pounds of laundry soap. The sample unit for testing shall be a bar of laundry soap. A 1-pound composite shall be prepared from the sample units randomly selected at the frequency listed below. The cutting, grinding and mixing for the preparation of the composite sample shall be conducted in an atmosphere of relative humidity within 5 percent of the relative humidity prevailing at time and place of weighing. The prepared composite shall be immediately placed in a clean, dry, airtight container marked with identifying lot number of date. Care shall be taken to prevent contamination or alteration of the composite during sampling, grinding, weighing, and storing. Duplicate determinations shall be made on the composite. The average of the two determinations shall comply with the requirements.

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE III. Instructions for testing of the end item

Characteristic	Specification reference: Requirement	Specification reference: Test method 1/	Requirement applicable to comp. samp.	Number of determinations	Results reported as
Moisture and matter volatile at 105°C	Table I	D-460 #11	X	2	Average of 2 determinations to nearest 0.1 percent.
Alkaline salts calculated as sodium oxide (Na_2O)	Table I	D-460 #20	X	2	Average of 2 determinations to nearest 0.1 percent.
Sum of free alkali or free acid, total matter insoluble in alcohol, and sodium chloride	Table I	D-460 #16-17	X	2	Average of 2 determinations to nearest 0.1 percent.
Free alkali calculated as sodium hydroxide (NaOH)	Table I	D-460 #18	X	2	Average of 2 determinations to nearest 0.1 percent.
Chloride calculated as sodium chloride (NaCl)	Table I	D-460 #50-52	X	2	Average of 2 determinations to nearest 0.1 percent.
Matter insoluble in water	Table I	D-460 #19	X	2	Average of 2 determinations to nearest 0.1 percent.
Anhydrous soap	Table I	D-460 #21-22	X	2	Average of 2 determinations to nearest 0.1 percent.

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TABLE III. Instructions for testing of the end item (Cont.)

Characteristic	Specification reference: Requirement	Specification reference: Test method 1/	Requirement applicable to comp. samp.	Number of determinations	Results reported as
Free acid, calculated as oleic acid	Table I	D-460 #18	X	2	Average of 2 determinations to nearest 0.1 percent.
Rosin (quantitative test)	Table I	D-460 #37-40	X	2	Average of 2 determinations to nearest 0.1 percent.
Titer mixed fatty acids prepared from the soaps	3.1	D-460 #43-44	X	2	Average of 2 determinations to nearest 1°C.

1/ All tests shall be performed in accordance with the designated Test Method.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The laundry soap bar does not require packaging for level A, B, or C.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Sixty size 1 or one hundred and twenty size 2 unwrapped bars of laundry soap shall be packed on end in a box conforming to PPP-B-636, grade V2s. Each box shall be provided with top and bottom pads made of the same material as the box. Size 1 bars of soap shall be arranged in 2 layers with 5 rows in each layer and 6 bars of soap per row. Size 2 bars of soap shall be arranged in such a manner that the box will have the same dimensions as the box for the size 1 bars of soap. Each shipping container shall be provided with a case liner constructed in accordance with type I of MIL-L-10547, except that the material shall consist of a lamination of 30-pound basis weight (24 by 36-500) kraft paper and 0.001-inch polyethylene. The bottom and side seams shall be heat sealed with seams not less than 3.8 inch in width. Closure of the case liner shall be accomplished by heat sealing or by a lock-seam, folded flat against the contents. Excess air shall be expelled prior to heat sealing of the closure. Each shipping container shall be closed, waterproofed and banded in accordance with the appendix of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Sixty (60) size 1 or one hundred and twenty (120) size 2 laundry soap bars shall be packed in a box conforming to PPP-B-636, class domestic. The box shall have a pad on the inside top and bottom of the same grade fiberboard as the box material. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.2 When waterproof containers are specified for level B shipments (see 6.2), laundry soap bars in the quantity and arrangement as specified in 5.2.1 shall be packed in a box conforming to PPP-B-636, class weather resistant, grade V3c or V3s or PPP-B-1163, grade 200. The box shall have a pad on the inside top and bottom of the same grade fiberboard as the box material. A case liner is not required. The box shall be closed and waterproofed in accordance with the appendix of the appropriate box specification. Banding of the V3c or V3s box is not required for level B packing.

5.2.3 Level C. Laundry soap of one size bar in quantities as specified (see 6.2.), shall be packed in a container complying with the Uniform Freight or National Motor Freight Classification, as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

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5.3 Marking.

5.3.1 Civil agencies. Shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The laundry soap covered by this specification is intended for use with soft to moderately hard water for hand dishwashing, cleaning of hard surfaces, and hand laundering where alkali salts are desirable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of the applicable level of packing (see 5.2).
(d) Whether waterproof containers are required for level B shipments (see 5.2.2.2).

6.3 Basis of purchase. The laundry soap shall be purchased by the case. The unit of purchase shall be one bar of soap.

Military coordinating activity:

Army - GL

Custodians:

Army - GL
Air Force - 84

Civil Agency Coordinating Activities:

VA - DMS
GSA - FSS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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